

Seed Signature Library for an In-House Signal and Noise Diagnostic Tool

Executive summary

A practical seed library for your tool should start with a **small, high-confidence core of signatures that are both diagnostically useful and externally defensible**. The strongest evidence base clusters into five families: power-quality disturbances defined by IEEE/IEC, switching and EMC signatures documented by IEC and industrial vendor notes, discrete-input timing failures documented by Rockwell/TI/ADI, rotating-machine faults supported by public bearing datasets, and measurement-artifact signatures documented by Tektronix and Rockwell. Those five families are enough to seed an internal library that is useful on day one without falling into scope creep. ¹

The most defensible first release is a registry of roughly **20–25 signatures**, each with: a short human description, time-domain and spectral markers, likely source buckets, best-fit analysis modules, confidence rules, suggested capture metadata, a synthetic generator recipe, and source references. Public datasets are strong for **power-quality** and **bearing fault** work, but much weaker for **industrial discrete-input and relay-noise cases**, where vendor application notes and standards are the better seed sources. That means your validation strategy should combine **dataset replay** for PQ and machinery with **synthetic signal tests** and **bench captures** for industrial control signatures. ²

For capture design, the key implementation rule is: **sample rate and record length must be chosen together**. NI's guidance on Nyquist and aliasing supports using sample rates above twice the highest frequency of interest, and Tektronix notes that single-shot transient fidelity is limited by sample rate while captured time span is set by record length. In practice, for fault diagnostics, using **~5× the highest frequency of interest as a default floor**, then increasing further for fast edges and short transients, is a sound starting rule rather than a fixed standard requirement. ³

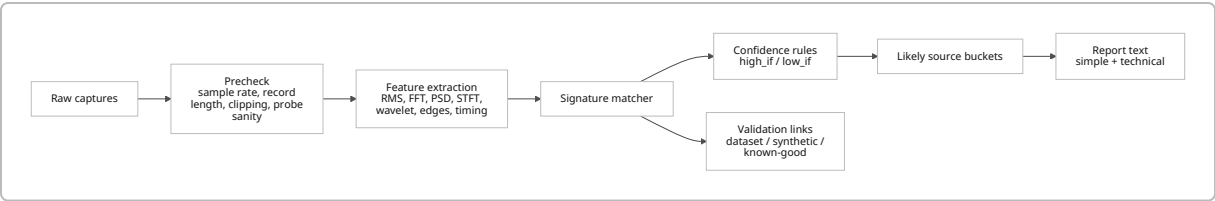
Evidence base and library design

The standards backbone is straightforward. IEEE Std 1159 is the most useful public anchor for naming and classifying common power-quality disturbances such as sags, swells, interruptions, impulsive transients, oscillatory transients, and notching. IEC 61000-4-30 defines repeatable in-situ power-quality measurement methods, while IEC 61000-4-7, -4-15, -4-11, -4-4, and -4-5 cover harmonics/interharmonics, flicker, dips and interruptions, EFT/burst, and surge respectively. IEC pages do not expose the full standard text publicly, so the library should use them mainly for **scope, terminology, and measurement intent**, with vendor notes and academic papers filling in observable signatures and capture practice. ⁴

For analysis modules, the mapping is stable. FFT and harmonic bins are appropriate for steady periodic content; STFT and wavelets are better for short, nonstationary events; coherence and cross-power spectrum are best for proving source-victim linkage; and impulse-response style reasoning is useful when you are

comparing how a path responds to an excitation. NI's docs explicitly tie coherence to the quality of the response-vs-stimulus relationship and define cross-power as the complex building block for related measurements. 5

For your in-house registry, the cleanest data model is a **signature entry** linked to one or more **source buckets**, one or more **analysis modules**, and one or more **validation artifacts** such as public datasets, synthetic generators, or known-good captures. That keeps the library extensible without requiring a giant physics simulator in version one. 6



The source-reference IDs used in the YAML examples map to the cited source list in the final section. Capture metadata in the examples below are **engineering defaults**, not universal standards; exact values remain application-specific where the source material does not prescribe them. 7

Summary table of seed signatures

The table below summarizes a **22-entry seed set** that is realistic for a first internal release.

id	label	primary source	key analyses	confidence level
pq_voltage_sag	Voltage sag	IEEE 1159 + IEC 61000-4-30/4-11	Sliding RMS, FFT, STFT	High
pq_voltage_swell	Voltage swell	IEEE 1159 + IEC 61000-4-30	Sliding RMS, FFT, STFT	High
pq_short_interruption	Short interruption	IEEE 1159 + IEC 61000-4-11	Sliding RMS, duration, STFT	High
pq_harmonic_distortion	Harmonic distortion	IEC 61000-4-7 + THD guidance	FFT, harmonic bins, THD, PSD	High
pq_flicker_am_mod	Flicker / low-frequency amplitude modulation	IEC 61000-4-15	RMS trend, sidebands, STFT	High

id	label	primary source	key analyses	confidence level
pq_commutation_notch	Commutation notch	IEEE 1159 + IEC 61000-4-7	Edge detection, FFT, wavelet	Medium-High
pq_impulsive_transient	Impulsive transient / spike	IEEE 1159 + IEC 61000-4-5	Peak detect, wavelet, STFT	High
pq_oscillatory_transient	Oscillatory transient / ringing	IEEE 1159 + IEC 61000-4-5	STFT, wavelet, ring-frequency estimation	High
emi_eft_burst	EFT/burst	IEC 61000-4-4	Edge counting, STFT, wavelet	High
current_inrush	Inrush current	TE relay/arc notes + industrial practice	Cycle RMS, peak, STFT, I-V timing	Medium-High
switch_relay_contact_bounce	Switch / relay contact bounce	Analog Devices + Rockwell	Edge timing, runt/glitch, pulse-width stats	High
relay_coil_inductive_kick	Relay / solenoid inductive kick	TE + IEC 61000-4-4 context	Peak detect, edge trigger, A/B comparison	High
ground_loop_hum	Ground-loop hum	TI + Tektronix	FFT, coherence, differential A/B compare	High
common_mode_noise	Common-mode HF noise	TI + Tektronix + Rockwell	Differential subtraction, CSD, coherence	High
pwm_vfd_edge_coupled_noise	PWM/VFD edge-coupled noise	Rockwell + NI coherence	Cross-correlation, coherence, STFT	High

id	label	primary source	key analyses	confidence level
sensor_threshold_chatter	Threshold chatter	TI comparator guidance	Edge timing, analog-vs-digital compare	High
slow_edge_late_transition	Slow edge / late transition	TI Schmitt-trigger guidance	Rise/fall, timing histogram	High
missed_short_pulse	Missed short pulse / scan-limited event	Rockwell fast DI docs	Pulse width, scope-vs-controller compare	High
high_speed_input_bounce	HSC bounce / over-filter tradeoff	Rockwell HSC docs	Count stats, filter A/B compare	High
bearing_fault_vibration_envelope	Bearing fault in vibration	CWRU + Paderborn	Envelope FFT, PSD, wavelet, STFT	High
bearing_fault_current_signature	Bearing fault in motor current	Paderborn + MCSA literature	FFT, PSD, STFT, coherence	Medium-High
broken_rotor_bar_sidebands	Broken rotor bar sidebands	MCSA literature	High-resolution FFT, order tracking	High

This seed set is intentionally biased toward signatures that have either a standards definition, a strong vendor-note measurement recipe, or a public dataset that can be replayed for validation. That is the right bias for a first internal library because it minimizes ambiguous entries that are hard to explain to users and hard to regression-test later. ⁸

Example full YAML entries

The examples below use `source_refs` to keep the YAML readable in-chat. Those IDs map to the cited source list in the final section.

```
signature_id: pq_voltage_sag
human_label: Voltage sag
short_description: RMS voltage drops below nominal for 0.5 cycle to 1 minute,
often due to faults or large motor starts.
```

```

time_domain_features:
  - step-down in sliding RMS envelope
  - sinusoidal shape often preserved during event
  - may coincide with current inrush
spectral_features:
  - fundamental remains dominant
  - sidebands may appear if load is modulated
  - little HF content unless a switching transient is superimposed
likely_source_buckets:
  - utility_or_facility_power_event
  - motor_starting
  - upstream_fault
best_analysis_modules:
  - sliding_RMS
  - FFT
  - THD
  - STFT
confidence_rules:
  high_if:
    - event lasts >= 0.5 cycle and <= 1 minute
    - multiple voltage channels agree on timing
    - current channel shows simultaneous inrush or fault response
  low_if:
    - only single-sample excursion is seen
    - event appears on one measurement setup only
example_capture_metadata:
  sample_rate: "10 kS/s minimum for PQ logging; 100 kS/s+ if edge context
matters"
  record_length: ">= 10 cycles before and after event"
  channels_needed:
    - Vphase
    - optional Iphase
  trigger: "RMS undervoltage or external event marker"
synthetic_test_signal_recipe: "50/60 Hz sine multiplied by an envelope stepping
to 0.1-0.9 pu for 0.5 cycle to 1 minute; optionally add inrush current on a
second channel."
representative_public_datasets_or_papers:
  - S01
  - S02
  - S03
  - S22
  - S24
suggested_report_language:
  simple: "Supply voltage dipped during the event."
  technical: "Sliding-RMS analysis indicates a voltage sag consistent with IEEE/
IEC power-quality classes; the waveform remains largely fundamental-dominated."

```

```

signature_id: pq_harmonic_distortion
human_label: Harmonic distortion
short_description: Steady waveform distortion with energy at integer multiples
of the line fundamental.
time_domain_features:
  - flattened or peaked waveform shape
  - repetitive non-sinusoidal distortion
  - RMS may stay near nominal while shape changes
spectral_features:
  - persistent integer-order harmonics
  - elevated THD
  - current harmonics often stronger than voltage harmonics near the source
likely_source_buckets:
  - rectifier_or_SMPS
  - VFD_front_end
  - nonlinear_load
best_analysis_modules:
  - FFT
  - harmonic_bins
  - THD
  - PSD
confidence_rules:
  high_if:
    - harmonics stay on integer bins across repeated captures
    - THD is elevated and repeatable
    - source-side current is more distorted than bus voltage
  low_if:
    - only broadband smearing is present
    - spectrum changes mainly with window choice or record coherence
example_capture_metadata:
  sample_rate: ">= 20 kS/s for up to 9 kHz harmonic view"
  record_length: "10-200 line cycles"
  channels_needed:
    - V_and_or_I
  trigger: "free-run with line reference if available"
synthetic_test_signal_recipe: "Sum the fundamental with 3rd, 5th, 7th, and 11th
components; vary phase and amplitude to emulate different nonlinear loads."
representative_public_datasets_or_papers:
  - S02
  - S04
  - S08
  - S22
  - S24
suggested_report_language:
  simple: "The waveform is distorted by harmonics."
  technical: "Spectral analysis shows persistent integer-order harmonic content
and elevated THD consistent with a nonlinear load source."

```

```

signature_id: emi_eft_burst
human_label: Electrical fast transient burst
short_description: Repetitive fast spikes arriving in bursts, typically from
switching arcs and similar EMI mechanisms.
time_domain_features:
  - clustered spikes rather than one isolated spike
  - burst packets over microseconds to milliseconds
  - often aligned with contactor or relay activity
spectral_features:
  - broadband energy
  - dense short-window comb-like content
likely_source_buckets:
  - relay_or_contactor_arcing
  - inductive_load_interruption
  - EFT_style_conducted_EMI
best_analysis_modules:
  - edge_counting
  - STFT
  - wavelet
  - event_statistics
confidence_rules:
  high_if:
    - multiple spikes arrive in bursts
    - burst timing correlates with switching activity
  low_if:
    - only one isolated spike is present
    - slow contact bounce is present without HF content
example_capture_metadata:
  sample_rate: "20-100 MS/s preferred"
  record_length: "few milliseconds around burst train"
  channels_needed:
    - victim_line
    - switching_source_optional
    - optional_PE
  trigger: "glitch or burst-density trigger"
synthetic_test_signal_recipe: "Inject packets of narrow pulses at 5 kHz or 100
kHz burst spacing on top of the baseline waveform."
representative_public_datasets_or_papers:
  - S06
  - S14
  - S17
suggested_report_language:
  simple: "Fast spiking occurred in bursts, not as a single event."
  technical: "Observed disturbance matches EFT/burst behavior associated with
repetitive fast transients from switching activity."

```

```

signature_id: switch_relay_contact_bounce
human_label: Switch or relay contact bounce
short_description: Multiple rapid make-break transitions before a mechanical
contact settles.
time_domain_features:
  - cluster of digital toggles over a millisecond-scale window
  - unstable state before final settle
  - may create false counts or multiple events
spectral_features:
  - broadband content during transition
  - edge timing is more useful than long-record FFT
likely_source_buckets:
  - mechanical_switch
  - relay_contact
  - pressure_or_limit_switch
best_analysis_modules:
  - edge_detection
  - runt_or_glitch_trigger
  - pulse_width_statistics
  - cross_correlation_to_actuation_command
confidence_rules:
  high_if:
    - multiple transitions occur within a mechanical settling interval
    - extra transitions disappear with debouncing or filter changes
  low_if:
    - only one clean edge is present
    - oscillation continues too long and looks more like chatter
example_capture_metadata:
  sample_rate: "1-10 MS/s for logic edge timing"
  record_length: "10-100 ms around actuation"
  channels_needed:
    - contact_voltage
    - optional_coil_or_command
  trigger: "edge, runt, or width trigger"
synthetic_test_signal_recipe: "Generate a logic transition followed by a damped
series of alternating edges with shrinking spacings."
representative_public_datasets_or_papers:
  - S18
  - S16
  - S14
suggested_report_language:
  simple: "The contact did not switch cleanly; it bounced before settling."
  technical: "Edge timing shows multiple make-break transitions within the
mechanical settling interval, consistent with contact bounce."

```

```

signature_id: ground_loop_hum
human_label: Ground-loop hum
short_description:
Low-frequency interference from ground potential differences, usually at 50/60
Hz and harmonics.
time_domain_features:
  - sinusoidal hum riding on a quiet signal
  - stable or slowly varying ground-related offset
spectral_features:
  - strong 50/60 Hz line component
  - harmonics may be present
  - common-mode contribution often dominates
likely_source_buckets:
  - multiple_ground_returns
  - remote_ground_difference
  - instrument_reference_error
best_analysis_modules:
  - FFT
  - coherence
  - differential_AB_compare
  - RMS
confidence_rules:
  high_if:
    - hum falls significantly when using differential measurement
    - strong line-frequency content is present on a node that should be quiet
  low_if:
    - content persists after proper differential isolation
    - dominant frequency is unrelated to the power system
example_capture_metadata:
  sample_rate: "10 kS/s is usually sufficient"
  record_length: "0.5-5 s"
  channels_needed:
    - suspect_node
    - reference_ground_or_second_node
  trigger: "free-run"
synthetic_test_signal_recipe: "Add 50/60 Hz common-mode content and a small
harmonic set to two channels with different ground references."
representative_public_datasets_or_papers:
  - S11
  - S12
  - S14
suggested_report_language:
  simple: "The measurement shows line-related hum consistent with a ground
loop."
  technical: "Dominant line-frequency common-mode content and improved results
under differential referencing indicate a ground-loop mechanism."

```

```

signature_id: pwm_vfd_edge_coupled_noise
human_label: PWM or VFD edge-coupled noise
short_description: Fast inverter edges couple into nearby victim wiring; the
victim disturbance stays synchronized to PWM transitions.
time_domain_features:
  - spikes or ringing repeat at PWM edges
  - noise is continuous while the drive is active
  - victim disturbance disappears or changes materially when drive is disabled
spectral_features:
  - switching-frequency families and harmonics
  - broadband bursts localized at edge timing
likely_source_buckets:
  - VFD_motor_leads
  - PWM_inverter
  - poor_cable_segregation_or_shielding
best_analysis_modules:
  - cross_correlation
  - coherence
  - STFT
  - edge_synchronous_averaging
confidence_rules:
  high_if:
    - victim waveform remains synchronized to suspect PWM edges
    - rerouting or shielding changes noise amplitude materially
  low_if:
    - timing drifts relative to PWM carrier
    - noise persists with drive disabled
example_capture_metadata:
  sample_rate: "10-100 MS/s for edge detail"
  record_length: "tens of switching periods plus one longer corroboration
record"
  channels_needed:
    - victim_line
    - PWM_phase_phase_or_phase_ground
  trigger: "PWM edge or width trigger"
synthetic_test_signal_recipe: "Generate a PWM waveform and inject delayed
damped spikes onto a victim channel at each rising and falling edge."
representative_public_datasets_or_papers:
  - S14
  - S15
  - S10
suggested_report_language:
  simple: "The noise is synchronized to inverter switching edges."
  technical: "Victim-channel disturbances remain phase-locked to PWM edges,
strongly implicating edge-coupled VFD or inverter EMI."

```

```

signature_id: missed_short_pulse
human_label: Missed short pulse
short_description: A real electrical pulse exists, but standard scan or filter
settings do not reliably register it.
time_domain_features:
  - oscilloscope sees a valid narrow pulse
  - controller image bit misses it or misses some of them
  - dedicated latch or timestamp may still record it
spectral_features:
  - pulse-width problem rather than a steady-state spectral problem
likely_source_buckets:
  - controller_scan_limit
  - input_filter_too_long
  - fast_sensor_event
best_analysis_modules:
  - pulse_width_histogram
  - scope_vs_controller_compare
  - event_latch_review
confidence_rules:
  high_if:
    -
hardware_pulse_capture_or_timestamp_records_the_event_while_the_normal_bit
misses_it
  - measured width is below the effective scan or filter window
low_if:
  - width exceeds configured limits with margin
  - event is absent electrically as well as logically
example_capture_metadata:
  sample_rate: "1-20 MS/s for pulse-width measurements"
  record_length: "enough to catch the rare event with pretrigger"
  channels_needed:
    - sensor_pulse
    - controller_capture_or_status_optional
  trigger: "pulse-width, runt, or glitch"
synthetic_test_signal_recipe: "Generate 5-100 microsecond pulses with varying
repetition spacing and pass them through a software model of input filtering."
representative_public_datasets_or_papers:
  - S16
  - S11
suggested_report_language:
  simple: "A real pulse is present, but the control system is not reliably
seeing it."
  technical: "Measured pulse width is below the effective capture window of the
configured input path; pulse-capture or shorter filter settings are indicated."

```

```

signature_id: broken_rotor_bar_sidebands
human_label: Broken rotor bar sidebands
short_description:
Rotor cage damage creates characteristic sidebands around the supply fundamental
related to slip.
time_domain_features:
  - subtle current modulation
  - load dependence is common
  - torque pulsation may be present
spectral_features:
  - high-resolution sidebands around line frequency
  - spacing should match slip-related expectation
  - sideband level relative to the fundamental trends severity
likely_source_buckets:
  - induction_motor_rotor_cage_fault
best_analysis_modules:
  - high_resolution_FFT
  - order_tracking_if_speed_varies
  - trend_vs_load
confidence_rules:
  high_if:
    - sidebands persist and strengthen with load
    - spacing is consistent with slip-derived physics
  low_if:
    - spacing is inconsistent with slip or pole-pass reasoning
    - sidebands disappear on repeat capture under the same operating point
example_capture_metadata:
  sample_rate: "5-20 kS/s is often enough, but long records are essential for
fine frequency resolution"
  record_length: "seconds to tens of seconds at steady load"
  channels_needed:
    - phase_current
    - optional_speed_or_load
  trigger: "free-run steady-state"
synthetic_test_signal_recipe: "Create a line-frequency current waveform with
low-amplitude amplitude modulation at twice-slip frequency to generate
sidebands around the fundamental."
representative_public_datasets_or_papers:
  - S20
  - S21
suggested_report_language:
  simple: "The motor current shows sidebands associated with rotor bar damage."
  technical: "High-resolution MCSA indicates classical slip-related sidebands
around the supply fundamental, consistent with broken rotor bars."

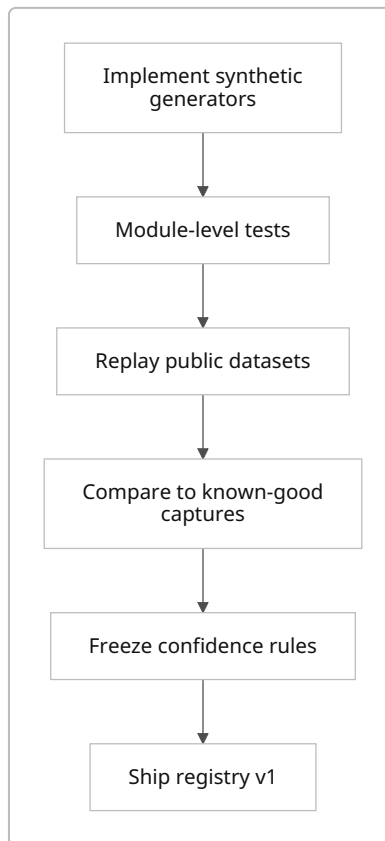
```

Unit tests and validation

The seed library should be validated with two complementary tracks: **synthetic unit tests** and **public-dataset replays**. Synthetic tests are mandatory because several important industrial signatures, especially relay bounce, coil kick, and scan-limited pulses, do not have strong public labeled datasets. Public replays are still invaluable for PQ and rotating machinery because they give you real noise, real operating variation, and known fault labels. ⁹

A sensible unit-test matrix is small and deterministic. For PQ, generate sag, swell, interruption, harmonics, flicker, notch, impulsive transient, oscillatory transient, and EFT burst from parameterized waveform models, then assert the correct module flags the correct labels and confidence rules. For industrial control, generate bounce trains, slow threshold crossings with and without hysteresis, narrow pulses below and above filter thresholds, and coil turn-off spikes with different suppression networks. For causal source-victim work, generate a source pulse train, make a delayed filtered copy on a victim channel, then assert that coherence and cross-correlation peak where expected. For machinery, replay CWRU and Paderborn records and assert that the healthy-vs-fault ranking changes in the expected direction for envelope FFT or MCSA features. ¹⁰

A concise validation plan is:



Recommended pass criteria for version one are conservative: the top-1 label should be correct on synthetic tests for the intended signature family; PQ classifier confidence should remain calibrated across the open-source PQ generators; machine-fault entries should separate healthy from damaged states on at least one

public dataset; and industrial-control entries should reproduce the expected behavior on a small internal bench setup. The goal for v1 is **reliable triage**, not universal diagnosis. ¹¹

Prioritized sources and limitations

The following prioritized sources are the best anchors for a first seed library, because they are either official standards pages, vendor documents with concrete measurement guidance, or public datasets and primary papers.

Standards and standards-adjacent anchors

S01. **IEEE Std 1159-2019** overview for power-quality disturbance classes, including transients, sags, swells, and interruptions. ¹²

S02. **IEC 61000-4-30** official scope for repeatable in-situ PQ measurement methods and the parameter family it standardizes. ¹³

S03. **IEC 61000-4-11** official scope for voltage dips, short interruptions, and voltage variations. ¹⁴

S04. **IEC 61000-4-7** scope and preview text for harmonics, interharmonics, instrumentation, and DFT-based measurement up to 9 kHz. ¹⁵

S05. **IEC 61000-4-15** official scope for flickermeter design and flicker severity evaluation. ¹⁶

S06. **IEC 61000-4-4** official scope for repetitive electrical fast transient/burst immunity testing. ¹⁷

S07. **IEC 61000-4-5** official scope for surge immunity, switching/lightning surges, and standard test intent. ¹⁸

Signal-processing and measurement-practice anchors

S08. **NI FFT and windowing** for FFT interpretation and spectral leakage reduction using windows. ¹⁹

S09. **NI sampling, Nyquist, and aliasing** for sample-rate guidance and alias prevention. ²⁰

S10. **NI coherence, cross-power spectrum, and impulse response** for source-response correlation, CSD usage, and time-domain interpretation of FRFs. ²¹

S11. **Tektronix ABCs of Probes** for ground-loop noise, short ground leads, probe-induced ringing, and differential best practice. ²²

S12. **TI common-mode noise survey** for ground loops, common-mode vs differential noise, and remote-ground problems. ²³

S13. **TI comparator chatter and Schmitt-trigger guidance** for hysteresis, slow-edge issues, and threshold chatter. ²⁴

Industrial control and EMI anchors

S14. **Rockwell System Design for Control of Electrical Noise** for source-victim segregation, measurement artifacts, PWM edge synchronization, intermittent-noise strategy, and differential measurements. ²⁵

S15. **Rockwell PWM drive wiring and grounding guide** for loop avoidance, routing, and why pigtail shield terminations are weak for noise-sensitive signals. ²⁶

S16. **Rockwell high-speed input and fast digital I/O docs** for debounce/filter tradeoffs, short-pulse detection, and configurable filter times. ²⁷

S17. **TE relay coil suppression note** for coil kick, suppression choices, release-dynamics effects, and failure modes. ²⁸

S18. **Analog Devices relay/switch bounce note** for contact bounce behavior, stress under inductive/capacitive loads, and debouncing patterns. ²⁹

Public datasets and academic seeds

S19. **Case Western Reserve University Bearing Data Center** for a canonical public bearing vibration dataset. ³⁰

S20. **Paderborn Bearing Data Center and associated PHM paper** for synchronized vibration and motor-current bearing-fault data with fault-state labels. ³¹

S21. **MCSA rotor-bar literature** for slip-related sidebands and practical broken-bar interpretation. ³²

S22. **Machlev open-source PQ disturbance generator** for synthetic PQ benchmarking. ³³

S23. **PQ-SyDa** for additional synthetic PQ event generation. ³⁴

S24. **Mendeley mathematical model of PQ disturbances** for a broader class list including sag, swell, interruption, notch, impulse, oscillatory transient, and combinations. ³⁵

S25. **Review of public PQ datasets** showing dataset coverage gaps and why synthetic PQ sets are still necessary. ³⁶

S26. **Recent STFT/wavelet-oriented PQ paper** supporting time-frequency methods for nonstationary disturbances. ³⁷

Open questions and limitations remain. Exact capture settings for your production tool are **not fully specified by the request** and depend on channel bandwidth, available scope memory, and the fastest event you intend to diagnose. Public labeled datasets are **weak for relay bounce, coil-kick, shield-termination, and scan-limited digital input faults**, so those entries should be validated primarily with synthetic generators and a small internal bench rig rather than by pretending a public benchmark exists. Finally, standard pages are authoritative for scope and taxonomy, but many detailed numeric limits live behind paywalls; where the public source only exposed scope text, I have kept the registry language deliberately conservative. ³⁸

¹ ⁴ ⁸ ¹² <https://ieeexplore.ieee.org/iel7/8796484/8796485/08796486.pdf>
<https://ieeexplore.ieee.org/iel7/8796484/8796485/08796486.pdf>

² ⁹ ³⁶ ³⁸ https://zaguan.unizar.es/record/96073/files/texto_completo.pdf
https://zaguan.unizar.es/record/96073/files/texto_completo.pdf

³ ⁷ ²⁰ <https://www.ni.com/en/shop/data-acquisition/measurement-fundamentals/analog-fundamentals/acquiring-an-analog-signal--bandwidth--nyquist-sampling-theorem-.html?srsId=AfmBOorktCr1B1haGfwP6JT2vj-yPZLcmjftjLkobj1aCM1GExFTV3z>
<https://www.ni.com/en/shop/data-acquisition/measurement-fundamentals/analog-fundamentals/acquiring-an-analog-signal--bandwidth--nyquist-sampling-theorem-.html?srsId=AfmBOorktCr1B1haGfwP6JT2vj-yPZLcmjftjLkobj1aCM1GExFTV3z>

⁵ https://www.ni.com/docs/en-US/bundle/labwindows-cvi/page/advancedanalysisconcepts/windowing_fa.html?srsId=AfmBOoqxjXPK8hx_B51U3q7Cx8yCY1IxfNNa6xzjuxHpS3mUWceU6Pen
https://www.ni.com/docs/en-US/bundle/labwindows-cvi/page/advancedanalysisconcepts/windowing_fa.html?srsId=AfmBOoqxjXPK8hx_B51U3q7Cx8yCY1IxfNNa6xzjuxHpS3mUWceU6Pen

⁶ ¹⁰ ¹¹ ³³ https://www.ipstconf.org/papers/Proc_IPST2021/21IPST093.pdf
https://www.ipstconf.org/papers/Proc_IPST2021/21IPST093.pdf

- 13 <https://webstore.iec.ch/en/publication/71611>
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